

LEONARD COURT

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EDUCATION

ETH Zurich, Switzerland	Expected Aug 2028
Master of Science in Robotics, Systems and Control	
University of Illinois at Urbana-Champaign (UofI)	2025 - 2026
Study Abroad Program in Mechanical Engineering	
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland	2023 - 2026
Bachelor of Science in Mechanical Engineering	GPA : 5.5/6

Related coursework :

Robot Dynamics	Probabilistic Artificial Intelligence	Model Predictive Control	Vision Algorithm	DPOC
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TECHNICAL SKILLS

Programming Languages : C++, C, Python, Matlab

Softwares/Tools : PyTorch, ROS 2, Docker, Eigen, Solidworks, Fusion 360, Abaqus, Catia

Spoken languages : French C2(Native), English C1 (Advanced), Spanish B1 (Intermediate)

WORK EXPERIENCE

Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland	Aug 2023 - Jul 2026
Teaching Assistant	

- Supported learning for cohort of 500+ students by explaining key concepts and problem-solving methods
- Guided students through exercises and provided strategies for academic success, graded mid-term tests and offered feedback
- Cours : *Physics: Mechanics (PHYS 101)* (Prof. Philippe Müllhaupt), *Physics: Thermodynamics (PHYS 106)* (Prof. Sylvain Bréchet), *Physics: Thermodynamics (PHYS 106)*(Prof. Jérémy Genoud), *Information, Computation, Communication (CS 119)*(Prof. Olivier Lévéque)

PROJECT HIGHLIGHTS

Learning-Based Robot Navigation: Expert vs. Imitation vs. RL (PyTorch, Stable-Baselines3, ROS 2, Docker)	Jul 2026 - Sep 2026
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- Built a 2D robot navigation benchmark and a hand-written VFH controller (97% success, 0 collisions), used as reference and demonstration source for imitation learning
- Designed a permutation-invariant DeepSets policy in PyTorch: 93% success vs. 33% for a flat MLP; DAgger reached expert level (96%) and cut collisions from 3% to 0.2%
- Trained PPO with reward shaping: matched the expert (97% on 500 unseen maps) without demonstrations while 34% faster; deployed as a ROS 2 node with Docker, unit tests and CI

EPFL Rocket Team – Firehorn 2025 (Ground Station Subsystem Engineer)	Sep 2024 - Jul 2026
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- Contributed to mechanical design, prototyping, and testing of ground support components for competition-level rocketry
- Developed CAD models and manufactured prototype parts using SolidWorks, 3D printing, CNC machining, and assembly
- Supported system integration between mechanical, electrical, and operational subsystems through testing and troubleshooting
- Collaborated in a multidisciplinary engineering team, applying problem-solving and design iteration to deliver practical solutions

Candy Crush (C Programming Project)

- Developed the full Candy Crush game in C and implemented AI capable of playing autonomously

School support platform, 'The Academy'	Jan 2021 - Jul 2023
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- Created and led online academic support platform bringing together several thousand students
- Organized peer-to-peer learning opportunities and student-led classes
- Collaborated with private advertisers, generating revenue through marketing partnerships

INTERESTS, ACTIVITIES & LEADERSHIP

EPFL Coaching : Student integration & mentoring	Sep 2024 - Jul 2025
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Sports : Practiced gymnastics at national level, Surf, Kite-Surf, Wing Foil, Wind-Surf